

Organic Chemistry, Carbs, and Lipids Study Guide

B1.1.1— Chemical properties of a carbon atom allowing for the formation of diverse compounds upon which life is based. *Students should understand the nature of a covalent bond. Students should also understand that a carbon atom can form up to four single bonds or a combination of single and double bonds with other carbon atoms or atoms of other non-metallic elements. Include among the diversity of carbon compounds examples of molecules with branched or unbranched chains and single or multiple rings.*

- Outline the number and type of bonds carbon can form with other atoms.

Organic: 4 stable nonpolar covalent with others

Single bonds, Single + double bonds, chains (branched or unbranched), rings (single or multiple)

- Recognize common functional groups.

Hydroxyl: -OH

Carboxyl: -C=O -OH

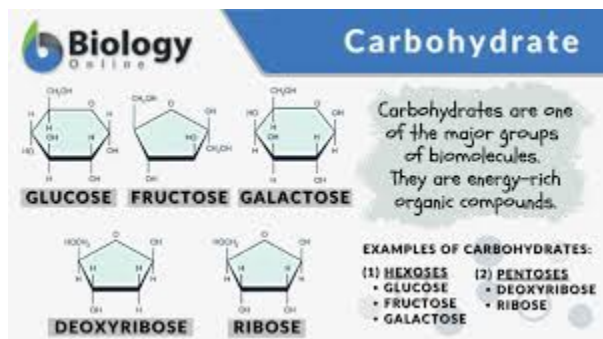
Amine: -NH₂

Phosphate: -O-P=O-O-O

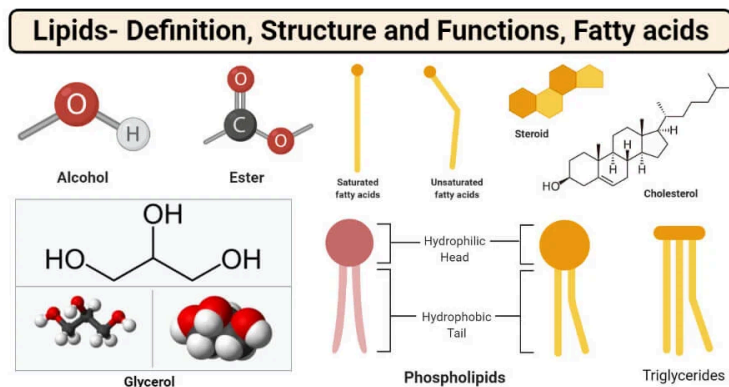
- List the four major classes of carbon compounds used by living organisms.

Carbohydrates: provide cells with quick, short-term energy

- monosaccharides

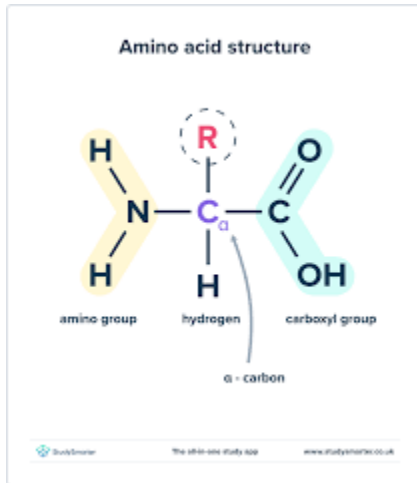


Lipids: provide cells with long-term energy + make up biological membranes



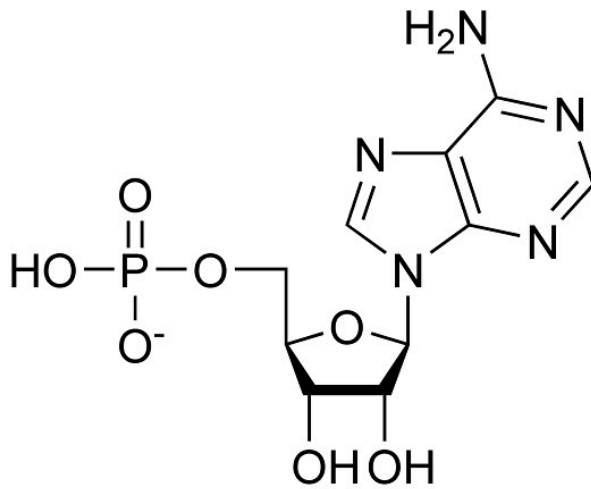
Proteins: Provide cell structure, hormones, speed up reaction

- amino acids

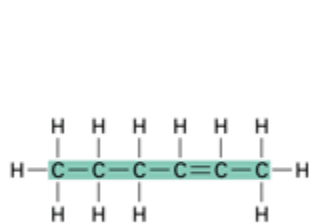


Nucleic acid: Store/pass on DNA/RNA

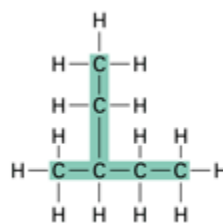
- nucleotide



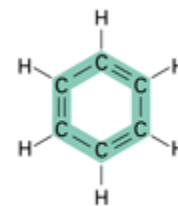
- List example molecules with branched chain, unbranched chain, single ring or multiple rings.



Straight chain



Branched chain



Ring

C1.1.3— Anabolic and catabolic reactions. *Examples of anabolism should include the formation of macromolecules from monomers by condensation reactions including protein synthesis (future unit), glycogen formation and photosynthesis (future unit). Examples of catabolism should include hydrolysis of macromolecules into monomers in digestion (future unit) and oxidation of substrates in respiration (future unit).*

- Contrast anabolic and catabolic reactions.

Anabolic: build up molecules

Catabolic: break down molecules

- List three examples of anabolic processes.

When condensation happens:

Glycogen formation

Photosynthesis

Protein synthesis: when amino acids come together to build a protein

- List three examples of catabolic processes.

When Hydrolysis happens:

Digestion, macromolecules into monomer

Oxidation of substrates in respiration

B1.1.2— Production of macromolecules by condensation reactions that link monomers to form a polymer. *Students should be familiar with examples of polysaccharides, polypeptides and nucleic acids.*

- Define monomer and polymer.

Monomer: subunits

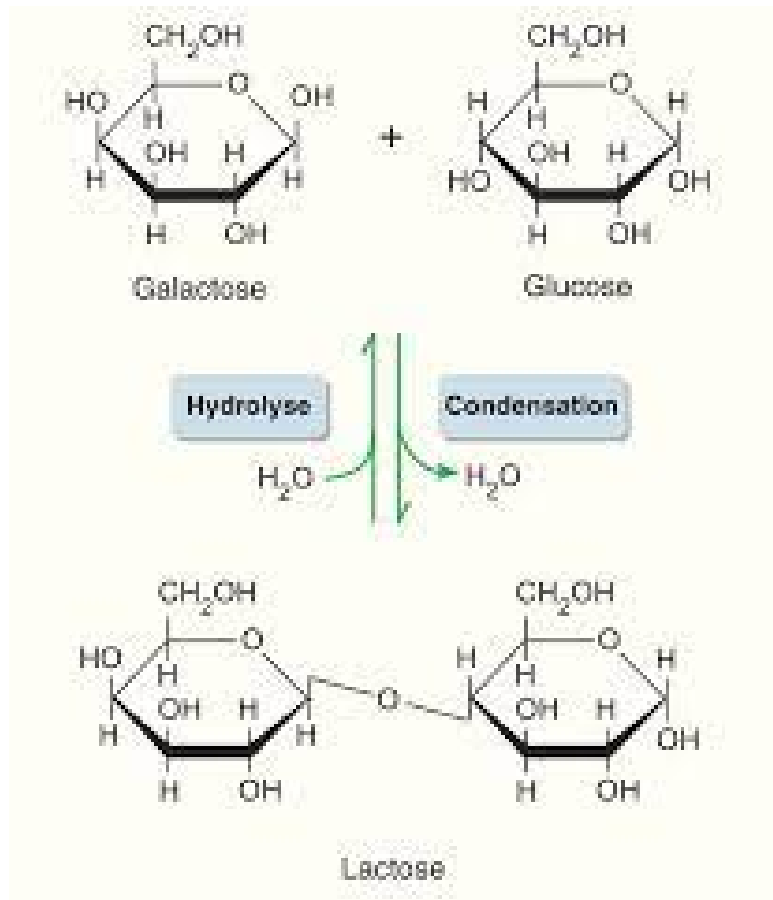
Polymer: the chain of monomers (subunits)

- Describe condensation reactions.

When water is a product

Condensation reactions join monomers together, forming a covalent bond and releasing water

- State that energy from ATP is needed to produce macromolecules by condensation reactions
 - Outline the condensation reactions that form polysaccharides, polypeptides and nucleic acids
- H and -OH are removed from adjacent monomers forming water and a bond



B1.1.3— Digestion of polymers into monomers by hydrolysis reactions. *Water molecules are split to provide the -H and -OH groups that are incorporated to produce monomers, hence the name of this type of reaction.*

- Define monomer and polymer.

Monomer- single building block

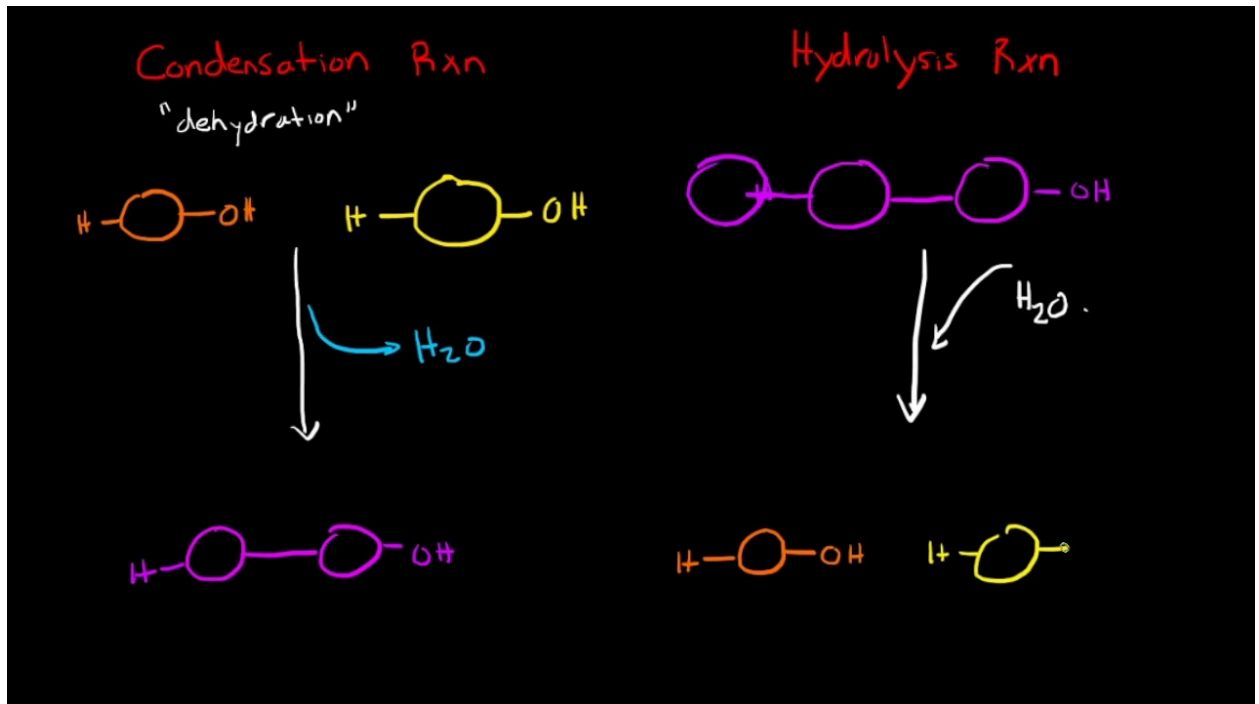
Polymer- multiple monomers

- Describe hydrolysis reactions.

When water is used to break up organic molecules, adding an -H and -OH

- Outline the hydrolysis reactions that digest polysaccharides, polypeptides and nucleic acids.

Water is being used to break apart one to make two



B1.1.4— Form and function of monosaccharides. Students should be able to recognize pentoses and hexoses as monosaccharides from molecular diagrams showing them in the ring forms. Use glucose as an example of the link between the properties of a monosaccharide and how it is used, emphasizing solubility, transportability, chemical stability and the yield of energy from oxidation as properties.

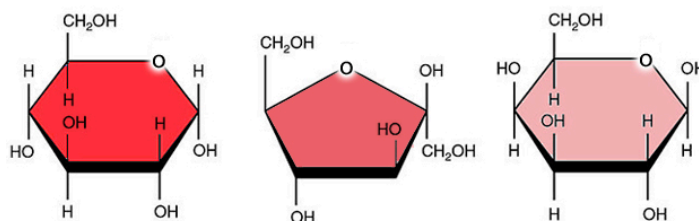
- Define monosaccharide.

The monomer for carbohydrates

5 or 6 ring

Only CHO (carbon hydrogen oxygen, 1:2:1 ratio)

Monosaccharides



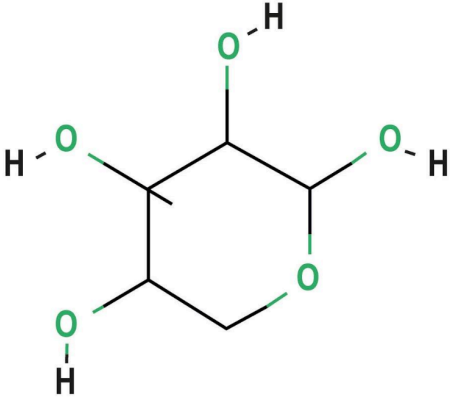
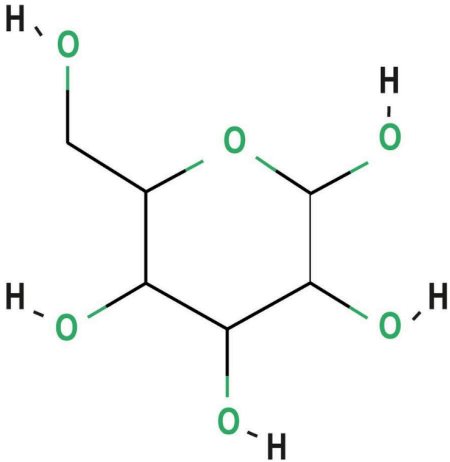
Glucose

Fructose

Galactose

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- Identify pentose and hexose carbohydrates from molecular diagrams.

	
Pentose	Hexose

- Outline the properties of glucose referring to solubility, transportability, stability, and energy yield from oxidation.

Is soluble, making it transportable in blood

Stable, useful for food storage, but if too much then is converted to glycogen or starch

Yield energy when oxidized, therefore used as substrate for respiration

B1.1.5— Polysaccharides as energy storage compounds. *Include the compact nature of starch in plants and glycogen in animals due to coiling and branching during polymerization, the relative insolubility of these compounds due to large molecular size and the relative ease of adding or removing alpha-glucose monomers by condensation and hydrolysis to build or mobilize energy stores.*

- Define polysaccharide.

Polymers of monosaccharides

Little energy to build

Easily reversible = release energy when digested via hydrolysis reaction

Starch + glucose are polysaccharides

- Compare the structure and function of amylose, amylopectin, and glycogen.

Amylose: plant, α-glucose, no branches, loops, coiled

Amylopectin: plant, α-glucose, branches, coral, branched glucose

Glycogen: animal, α-glucose, many branches

- Discuss the benefit of polysaccharide coiling and branching during polymerization.

Allows for more compact storage of starch and glycogen

- Explain how condensation or hydrolysis of α-glucose monomers build or mobilize energy stores.

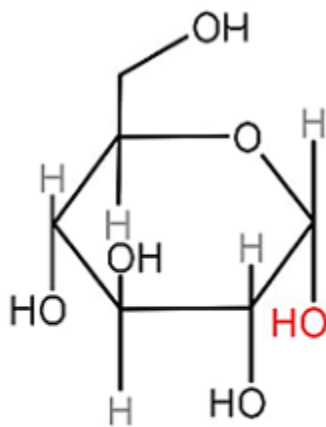
Condensation builds up to make glucose and glycogen

B1.1.6- Structure of cellulose related to its function as a structural polysaccharide in plants. *Include the alternating orientation of beta-glucose monomers, giving straight chains that can be grouped in bundles and cross-linked with hydrogen bonds.*

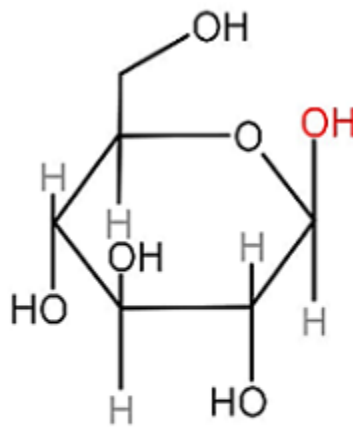
- Compare the structure of α-glucose and β-glucose.

Alpha glucose has H and OH on one side the same

Beta glucose has H and OH alternating



Alpha-glucose



Beta-glucose

KingDraw

- Describe the structure of cellulose microfibrils.

Elongated structure in plant cell walls composed of cellulose chains in a crystalized fashion

- Discuss the consequences of the strength of cellulose in the plant cell wall.

Maintains the shape of the plant

B1.1.8- Hydrophobic properties of lipids. *Lipids are substances in living organisms that dissolve in non-polar solvents but are only sparingly soluble in aqueous solvents. Lipids include fats, oils, waxes and steroids.*

- Explain why lipids are hydrophobic.

They are nonpolar

- Outline the structure and function of fats, oils, waxes and steroids.

They are all a type of lipids

Fats: to serve as energy storage

Oils: to serve as energy storage

Waxes: layer against water found in top layer of plants

Steroids: hormone regulation, different functioned groups to rings = different function

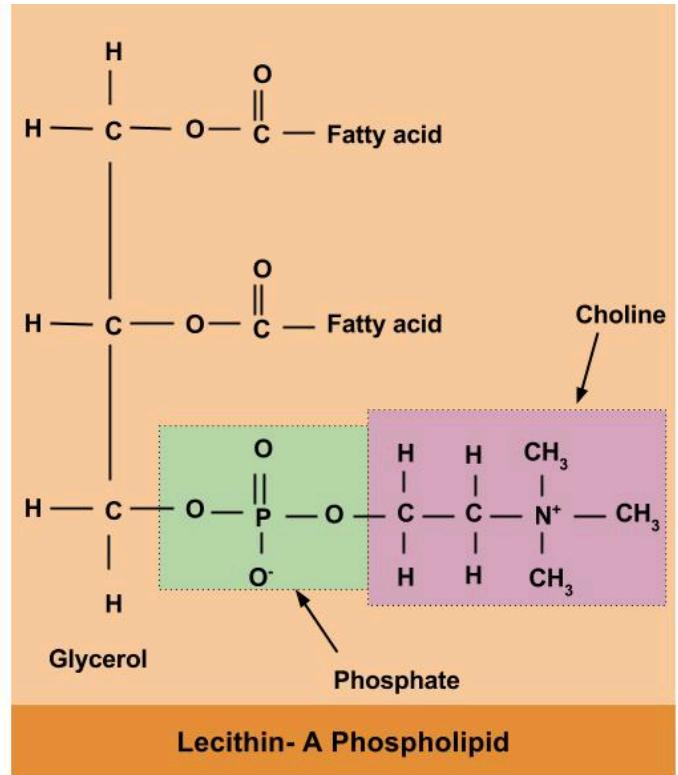
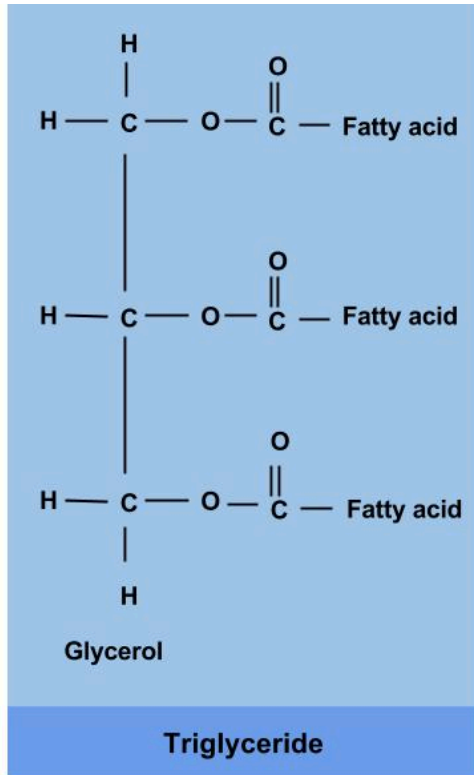
B1.1.9- Formation of triglycerides and phospholipids by condensation reactions. *One glycerol molecule can link three fatty acid molecules or two fatty acid molecules and one phosphate group.*

- Explain the condensation reaction connecting fatty acids and glycerol to form a triglyceride.

Formed by condensation: 3 fatty acids + glycerol, creating ester bonds between glycerol and fatty acid

- Explain the condensation reaction connecting fatty acids, glycerol and a phosphate group to form a phospholipid.

Formed by 2 fatty acids + phosphate group + glycerol → amphipathic molecule



B1.1.10- Difference between saturated, monounsaturated and polyunsaturated fatty acids. *Include the number of double carbon (C=C) bonds and how this affects melting point. Relate this to the prevalence of different types of fatty acids in oils and fats used for energy storage in plants and endotherms respectively.*

- Describe the structure of a generalized fatty acid.

Long, straight chain of carbon atoms with hydrogen in each with a (carboxyl) COOH at the end

- Compare and contrast the structures and properties of saturated and unsaturated (mono- or poly-) fatty acids.

Saturated: all possible H are there, no C=C bonds meaning higher melting point

Unsaturated: not all possible H are there, one or more C=C bonds meaning lower melting point

- Distinguish between the structure and properties of cis- and trans-unsaturated fatty acids.

Cis: on the same side → bent chain

Trans: hydrogens opposite (up and down), every other C, stronger chain

B1.1.11- Triglycerides in adipose tissues for energy storage and thermal insulation. *Students should understand that the properties of triglycerides make them suited to long-term energy storage functions. Students should be able to relate the use of triglycerides as thermal insulators to body temperature and habitat.*

- Outline properties of triglycerides that make them suitable for long-term energy storage.

Chemically stable, energy not lost over time

Immiscible with water,

Release twice as much energy in cell respiration as carbohydrate

Poor conductors of heat, can be used as thermal insulator to conserve body heat

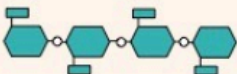
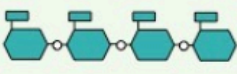
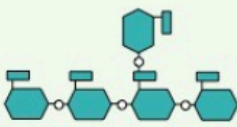
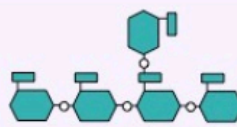
Liquid at body temperature, can act as a shock absorber acting as insulation

- State the function of adipose tissue.

Located beneath skin and some organs

- Discuss the adaptation of a thick adipose tissue layer as a thermal insulator.

Low thermal conductivity, so conserves heat

	Cellulose	Starch		Glycogen
		Amylose	Amylopectin	
Source	Plant	Plant	Plant	Animal
Subunit	β -glucose	α -glucose	α -glucose	α -glucose
Branches	No	No	Yes	Yes
Diagram				
Shape	